



DIBBs eCR Viewer Overview

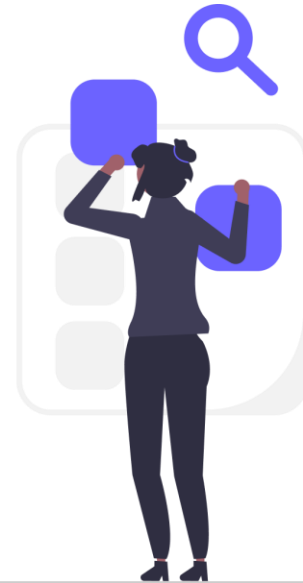
Data Integration Building Blocks (DIBBs) Team

Challenges & Solution

The Challenge: Complex Data

It's difficult for case investigators, epis, and others to **find the information they need** in an eCR.

Users need to **search through lengthy documents** for information relevant to a case.



Our Solution: eCR Viewer

Provides a **human-readable** view of eCRs.

Designed for **case investigators**, **eCR coordinators**, and others who review individual eCRs.

The screenshot displays the 'eCR Viewer' interface. At the top, a dark blue header contains the title 'eCR Viewer'. Below it, a teal bar shows the patient name 'Bugs Bunny' and the date '12/07/1989'. A navigation bar includes a back arrow and the link 'Back to eCR Library'. On the left, a sidebar lists various sections: 'eCR Summary' (selected), 'eCR Document', 'Patient Info' (with sub-items: Demographics, Social History), 'Encounter Info' (with sub-items: Encounter Details, Facility Details, Provider Details), 'Clinical Info' (with sub-items: Symptoms and Problems, Treatment Details, Immunizations, Diagnostics and Vital Signs), 'Lab Info' (with sub-item: Lab Results from Castle Lab), 'eCR Metadata' (with sub-items: RR Details, eCR Details), 'eCR Author Details for Practitioner', 'eCR Custodian Details', and 'Unavailable Info'. The main content area is titled 'eCR Summary' and includes a subtitle 'Provides key info upfront to help you understand the eCR at a glance'. It features two primary sections: 'Patient Summary' and 'Encounter Summary'. The 'Patient Summary' section contains fields for Patient Name (Bugs Bunny), DOB (12/07/1989), Sex (Female), Race (White), Ethnicity (Not Hispanic or Latino), Patient Address (999 Single Lane, Hudson County, NY 07086, USA), Patient Contact (Home: +1-999-999-9999, Mobile: +1-999-999-9999, bugs.bunny@gmail.com), and Encounter Summary. The 'Encounter Summary' section includes Encounter Date/Time (Start: 02/18/2022 9:01 AM, End: 02/18/2022 10:47 AM), Encounter Type (Ambulatory, Transcribe Orders), Encounter Diagnosis (COVID-19), Encounter Reason (Dull chest pain), Facility Name (Castle Outpatient Clinic), and Facility Contact (Work: 1-999-999-9999). Below these is a 'Condition Summary' section with a link to '4 CONDITIONS FOUND'. It lists three conditions: 'Viral hepatitis type C (disorder)', 'Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)', and 'Plague (disorder)', each with a plus icon for expansion.

Patient Summary	
Patient Name	Bugs Bunny
DOB	12/07/1989
Sex	Female
Race	White
Ethnicity	Not Hispanic or Latino
Patient Address	999 Single Lane Hudson County, NY 07086, USA
Patient Contact	Home: +1-999-999-9999 Mobile: +1-999-999-9999 bugs.bunny@gmail.com

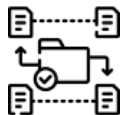
Encounter Summary	
Encounter Date/Time	Start: 02/18/2022 9:01 AM End: 02/18/2022 10:47 AM
Encounter Type	Ambulatory Transcribe Orders
Encounter Diagnosis	COVID-19
Encounter Reason	Dull chest pain
Facility Name	Castle Outpatient Clinic
Facility Contact	Work: 1-999-999-9999

Condition Summary 4 CONDITIONS FOUND	
Viral hepatitis type C (disorder)	+
Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	+
Plague (disorder)	+

eCR Viewer Features



Surfaces the most relevant information in a summary at the top of the eCR, so any staff member can quickly find the information they need



Combines information from both the eICR and RR, so epidemiologists reviewing the eCR can understand why it was sent to their jurisdiction



Keeps data formats consistent regardless of which EHR system generated the eCR

eCR Viewer

Bugs Bunny 12/07/1989

[Back to eCR Library](#)

eCR Summary

eCR Document

Patient Info

Demographics

Social History

Encounter Info

Encounter Details

Facility Details

Provider Details

Clinical Info

Symptoms and Problems

Treatment Details

Immunizations

Diagnostics and Vital Signs

Lab Info

Lab Results from Castle Lab

eCR Metadata

RR Details

eICR Details

eICR Author Details for Practitioner

eICR Custodian Details

Unavailable Info

eCR Summary

Provides key info upfront to help you understand the eCR at a glance

Patient Summary

Patient Name	Bugs Bunny
DOB	12/07/1989
Sex	Female
Race	White
Ethnicity	Not Hispanic or Latino
Patient Address	999 Single Lane Hudson County, NY 07086, USA
Patient Contact	Home: +1-999-999-9999 Mobile: +1-999-999-9999 bugs.bunny@gmail.com

Encounter Summary

Encounter Date/Time	Start: 02/18/2022 9:01 AM End: 02/18/2022 10:47 AM
Encounter Type	Ambulatory Transcribe Orders
Encounter Diagnosis	COVID-19
Encounter Reason	Dull chest pain
Facility Name	Castle Outpatient Clinic
Facility Contact	Work: 1-999-999-9999

Condition Summary 4 CONDITIONS FOUND

Viral hepatitis type C (disorder)	+
Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	+
Plague (disorder)	+

Surfaces Relevant Information

eCR SUMMARY

Provides critical information about an eCR at a glance

This includes:

- Patient summary
- Encounter summary
- Condition summary

eCR Summary

Provides key info upfront to help you understand the eCR at a glance

Patient Summary

Patient Name	Bugs Bunny
DOB	12/07/1989
Sex	Female
Race	White
Ethnicity	Home: +1-615-999-9999 Mobile: +1-615-999-9999 bugs.bunny@gmail.com
Patient Address	999 Single Lane Hudson County, NY 07086, USA
Patient Contact	Home: +1-999-999-9999 Mobile: +1-999-999-9999 bugs.bunny@gmail.com

Encounter Summary

Encounter Date/Time	Start: 02/18/2022 9:01 AM End: 02/18/2022 10:47 AM
Encounter Type	Ambulatory Transcribe Orders
Encounter Diagnosis	COVID-19
Encounter Reason	Dull chest pain
Facility Name	Castle Outpatient Clinic
Facility Contact	Work: 1-999-999-9999

Condition Summary 4 CONDITIONS FOUND

Viral hepatitis type C (disorder)	+
Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	+
Plague (disorder)	+
Anthrax (disorder)	-

Surfaces Relevant Information

CONDITION SUMMARY

Highlights relevant information to the reportable conditions

This includes:

- RCKMS rule summary
- Clinical Information Relevant to the Reportable Condition
- Labs Relevant to the Reportable Condition

Condition Summary

4 CONDITIONS FOUND

Viral hepatitis type C (disorder)

+

Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)

+

Plague (disorder)

+

Anthrax (disorder)

-

RCKMS Rule Summary

Anthrax vaccine administered

Immunizations Relevant to Reportable Condition

Name	Administration Dates	Dose Number	Manufacturer	Lot Number
Anthrax	02/30/2022	1	Pfizer	LOT9911Z

Lab Results Relevant to Reportable Condition

No matching lab results found in this eCR

Consistent Format

CONSISTENT FORMAT

Standardized layout of data, regardless of provider or triggering EHR

This includes:

- Side menu
- Clear sections
- Structured fields

Culture Bact CSF		ABNORMAL	02/17/2022 1:15 PM CDT	
Component	Value	Ref Range	Test Method	Lab Comment
Final Report	Light growth Acinetobacter baumannii	No data	No data	No data
Organism ID	Acinetobacter baumannii (POSITIVE)	No data	No data	No data
Gram Stain Report	1+ Gram negative rod	No data	No data	View comment
Organism	Antibiotic	Method	Susceptibility	Lab Comment
Acinetobacter baumannii	Amikacin	MIC	0.25: Susceptible	View comment
Acinetobacter baumannii	Avycaz (Ceftazidime/ Avibactam)	MIC	0.05: Intermediate	View comment
Acinetobacter baumannii	Levofloxacin	MIC	0.75: Resistant	View comment
Analysis Time		No data		
Collection Time		02/17/2022 1:15 PM CDT		
Received Time		02/17/2022 1:15 PM CDT		

Integrations

Two options offered:

1. Direct integration of core Viewer page within a surveillance system



Home | Data Entry | Merge Patients | Open Investigations | Reports | System Management

View Document

Create Investigation | Transfer Ownership | Delete

thorX1 odinsonX1 | Female | 04/20/1990 | Patient ID: 67084

Jurisdiction: Fulton County | Program Area: ARBO | Created: 02/24/2023 | By: PKS PKS

Date Received: 02/24/2023 | Document ID: DCC1000002GA51 (Update) | Last Updated: 02/24/2023 | By: PKS PKS

Public Health Case Report - Data From Legacy System To CDA

Go to: [SENDING SYSTEM INFORMATION](#) | [PATIENT INFORMATION](#) | [SOCIAL HISTORY INFORMATION](#) | [CLINICAL INFORMATION](#) | [Encounters](#) | [Social History](#) | [Plan of Treatment](#) | [Results](#) | [Problems](#) | [History of Present Illness](#) | [Medications](#)

[Administered](#) | [Research Use Only](#)

SENDING SYSTEM INFORMATION

Message Type	Public Health Case Report - PHRI	Sending System ID	2.16.840.1.113883.19.5
Sending Facility	Level Seven Healthcare, Inc.	Sending System	
Message Creation Date	November 7, 2016, 09:44:21	Message Status	Set: ONGOING_CASE Version: 1

[Back to top](#)

PATIENT INFORMATION

Patient Name	thorX1 odinsonX1	Sending System Patient ID	CSR100010X007 (2.16.840.1.113883.4.1) 111-00-1234 (2.16.840.1.114222.4.5.1)
Date of Birth	April 20, 1990	Reported Age	
Race	White (2106-3)	Ethnicity	Hispanic or Latino (2135-2)
Sex	Female (F)	Contact Info	Home: 21 Baker Street Atlanta, NC, MAPRED, Georgia 30306, United States +1-655-555-2003 (Primary Home)

[Back to top](#)

SOCIAL HISTORY INFORMATION

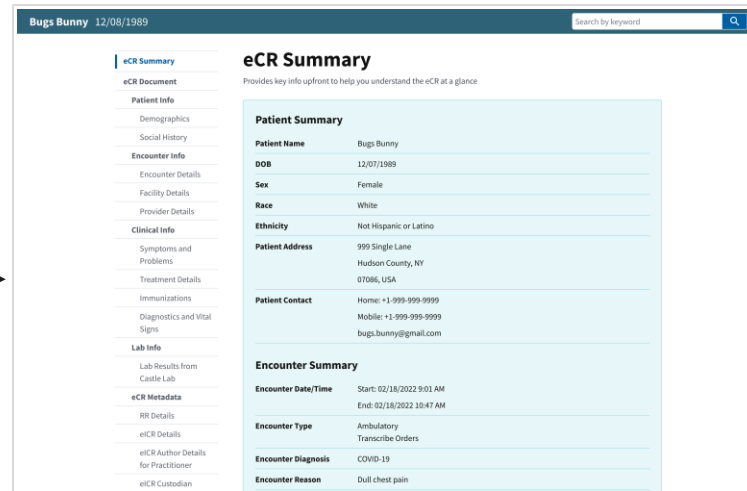
Birth Sex:	Female (F) / Female (F)	Is this person deceased?	No (N) / FALSE (false)
------------	-------------------------	--------------------------	------------------------

[Back to top](#)

CLINICAL INFORMATION

Investigation ID	2.16.840.1.113883.4.6.9037012.3928002	Condition	Zka virus disease (disorder) (3928002)
Pregnancy Status	No (N) / No (N)	Physician	Henry Seven 4444 Healthcare Drive Ann Arbor, Michigan 99999, United States

NBS - "View eICR Document" Button



Bugs Bunny 12/08/1989

Search by keyword

eCR Summary

Provides key info upfront to help you understand the eCR at a glance

Patient Summary

Patient Name	Bugs Bunny
DOB	12/07/1989
Sex	Female
Race	White
Ethnicity	Not Hispanic or Latino
Patient Address	999 Single Lane Hudson County, NY 07086, USA
Patient Contact	Home: +1-999-999-9999 Mobile: +1-999-999-9999 bugs.bunny@gmail.com

Encounter Summary

Encounter Date/Time	Start: 02/18/2022 9:01 AM End: 02/18/2022 10:47 AM
Encounter Type	Ambulatory Transcribe Orders
Encounter Diagnosis	COVID-19
Encounter Reason	Dull chest pain

Individual eCR

Integrations

Two options offered:

2. Standalone tool – the eCR Library

eCR Viewer				
eCR Library				
Search by patient or keyword				
FILTERS: Last 24 hours Reportable Condition				
Patient	Received Date	Encounter Date	Reportable Condition	RCKMS Rule Summary
Darth Maulie V3 DOB: 11/04/1989	12/23/2022 2:59 PM	12/23/2022 2:59 PM	Viral hepatitis type C (disorder)	Detection of Hepatitis C virus antibody in a clinical specimen by any method
			Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	Detection of SARS-CoV-2 nucleic acid in a clinical or post-mortem specimen by any method
				COVID-19 (as a diagnosis or active problem)
General Greivouss V2 DOB: 2/17/1984	12/23/2022 2:59 PM	12/23/2022 2:59 PM	Overdose of opiate (disorder)	Detection of opioid substance or metabolite in a clinical specimen by any method
				Vital status at end of encounter is alive or unknown
Luke Skywalker DOB: 12/12/2020	12/23/2022 1:43 PM	12/23/2022 1:43 PM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Captain Rex DOB: 4/3/2005	12/23/2022 10:57 AM	12/23/2022 10:57 AM	Salmonella infection (disorder)	Detection of Salmonella species and serovars nucleic acid (except S. typhi and S. paratyphi nucleic acid) in a clinical specimen by any method
Death Vader DOB: 1/14/1948	12/23/2022 9:00 AM	12/23/2022 9:00 AM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Count Dooku DOB: 7/22/1984	12/23/2022 8:05 AM	12/23/2022 8:05 AM	Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	COVID-19 (as a diagnosis or active problem)
Admiral Motti	12/23/2022	12/23/2022	Disease caused by severe acute respiratory	COVID-19 (as a diagnosis or active problem)
Showing 1-25 of 757 eCRs				
1 2 3 4 ... 38 Next >				eCRs per page 25

eCR Library

eCR Viewer	
Bugs Bunny 12/07/1989	
Back to eCR Library	
eCR Summary	
Provides key info upfront to help you understand the eCR at a glance	
eCR Document	
Patient Info	
Demographics	
Social History	
Encounter Info	
Encounter Details	
Facility Details	
Provider Details	
Clinical Info	
Symptoms and Problems	
Treatment Details	
Immunizations	
Diagnostics and Vital Signs	
Lab Info	
Lab Results from Castle Lab	
eCR Metadata	
RR Details	
Patient Summary	
Patient Name	Bugs Bunny
DOB	12/07/1989
Sex	Female
Race	White
Ethnicity	Not Hispanic or Latino
Patient Address	999 Single Lane Hudson County, NY 07086, USA
Patient Contact	Home: +1-999-999-9999 Mobile: +1-999-999-9999 bugs.bunny@gmail.com
Encounter Summary	
Encounter Date/Time	Start: 02/18/2022 9:01 AM End: 02/18/2022 10:47 AM
Encounter Type	Ambulatory

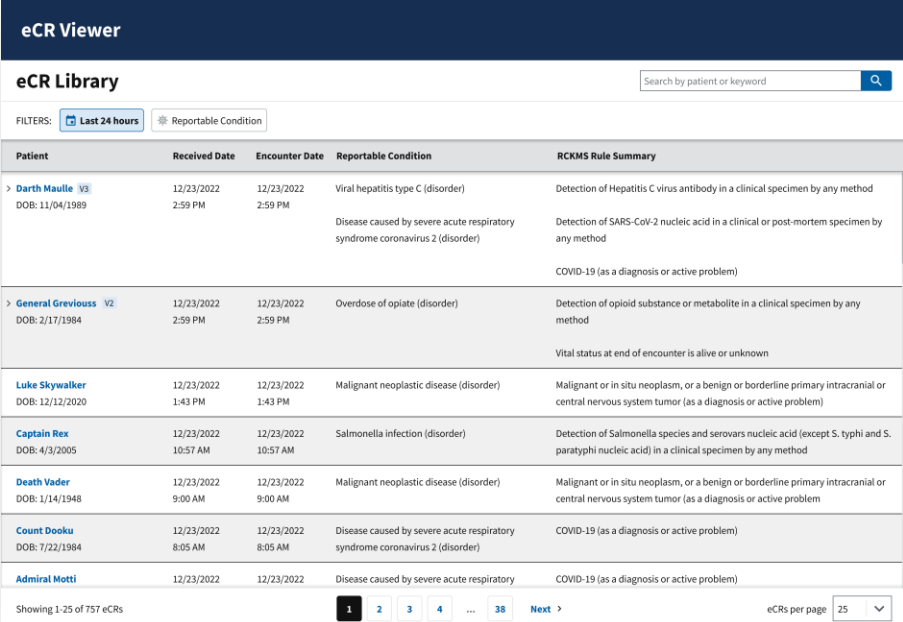
Individual eCR

Standalone Features

eCR Library offers:

- Searching by patient
- Grouping eCRs by encounter
- Filtering eCRs by date or reported condition

This version also creates a **database of parsed eICR/RR data** for public health departments to use directly.



The screenshot displays the 'eCR Viewer' application. At the top, there's a dark blue header with the title 'eCR Viewer'. Below it, a white section titled 'eCR Library' contains a search bar with the placeholder text 'Search by patient or keyword'. Under the search bar, there are filter buttons: 'Last 24 hours' (selected) and 'Reportable Condition'. The main content is a table with the following columns: Patient, Received Date, Encounter Date, Reportable Condition, and RCKMS Rule Summary. The table lists several eCRs for different patients, including Darth Maulie, General Greivous, Luke Skywalker, Captain Rex, Death Vader, Count Dooku, and Admiral Motti. Each row shows the patient's name and DOB, the received and encounter dates and times, the reportable condition, and the corresponding RCKMS rule summary. At the bottom of the table, it says 'Showing 1-25 of 757 eCRs'. To the right of this text is a pagination control with buttons for 1, 2, 3, 4, ..., 38, and a 'Next' button. Further to the right is a dropdown menu for 'eCRs per page' set to 25.

Patient	Received Date	Encounter Date	Reportable Condition	RCKMS Rule Summary
> Darth Maulie V3 DOB: 11/04/1989	12/23/2022 2:59 PM	12/23/2022 2:59 PM	Viral hepatitis type C (disorder) Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	Detection of Hepatitis C virus antibody in a clinical specimen by any method Detection of SARS-CoV-2 nucleic acid in a clinical or post-mortem specimen by any method COVID-19 (as a diagnosis or active problem)
> General Greivous V2 DOB: 2/17/1984	12/23/2022 2:59 PM	12/23/2022 2:59 PM	Overdose of opiate (disorder)	Detection of opioid substance or metabolite in a clinical specimen by any method Vital status at end of encounter is alive or unknown
Luke Skywalker DOB: 12/12/2020	12/23/2022 1:43 PM	12/23/2022 1:43 PM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Captain Rex DOB: 4/3/2005	12/23/2022 10:57 AM	12/23/2022 10:57 AM	Salmonella infection (disorder)	Detection of Salmonella species and serovars nucleic acid (except S. typhi and S. paratyphi nucleic acid) in a clinical specimen by any method
Death Vader DOB: 1/14/1948	12/23/2022 9:00 AM	12/23/2022 9:00 AM	Malignant neoplastic disease (disorder)	Malignant or in situ neoplasm, or a benign or borderline primary intracranial or central nervous system tumor (as a diagnosis or active problem)
Count Dooku DOB: 7/22/1984	12/23/2022 8:05 AM	12/23/2022 8:05 AM	Disease caused by severe acute respiratory syndrome coronavirus 2 (disorder)	COVID-19 (as a diagnosis or active problem)
Admiral Motti	12/23/2022	12/23/2022	Disease caused by severe acute respiratory	COVID-19 (as a diagnosis or active problem)

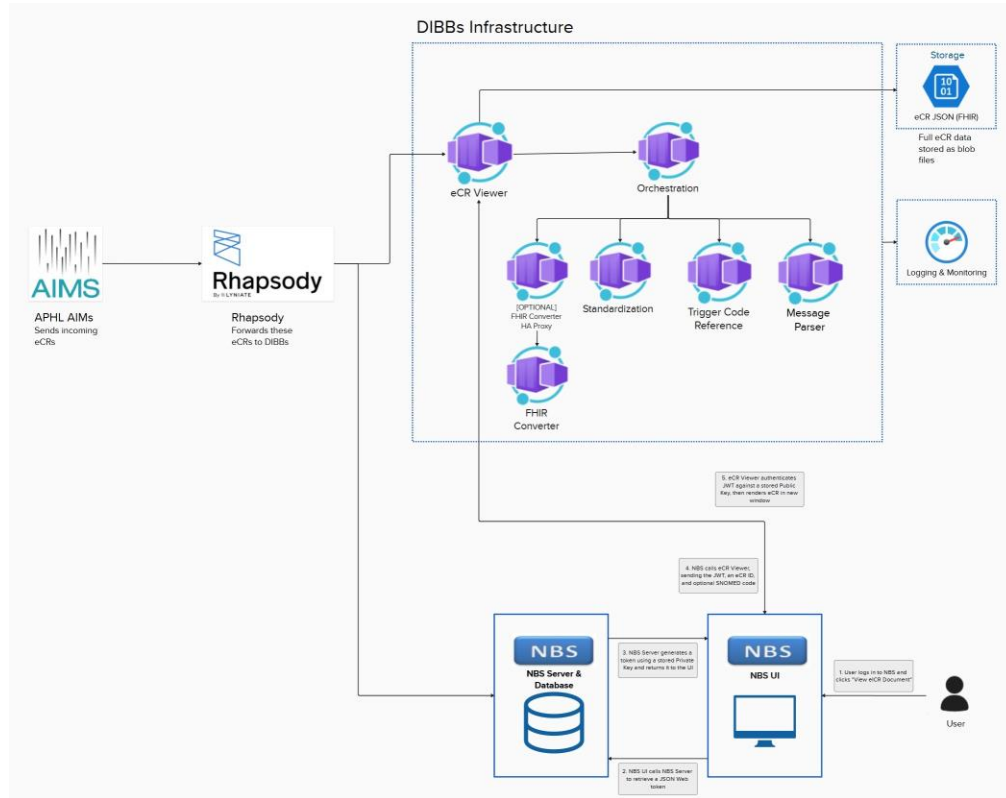
Showing 1-25 of 757 eCRs

1 2 3 4 ... 38 Next >

eCRs per page 25

Technical Overview

Technical Overview



- **Docker containers** for application code
- **Blob storage** for FHIR eCR/RR bundles
- **SQL database** (if using eCR Library)

Deployment Strategy

- Cloud-based deployment within your jurisdiction – **data never leaves the STLT environment**
- 3 deployment types directly supported:
 - [AWS Terraform package](#)
 - [Azure Terraform package](#)
 - Hybrid [virtual machine](#) (VM + cloud storage)
- 2 other user-created deployments:
 - Custom AWS Terraform (TN)
 - Kubernetes (NBS 7 team)



Adopting the Viewer

Prerequisites for eCR Viewer



eCR DATA

eCRs must be authored in RCKMS and available in your jurisdiction.



CLOUD ENVIRONMENT

The Viewer is compatible with all 3 major cloud providers (AWS, Azure, and GCP).



DEPLOYMENT TEAM

An internal team to deploy and manage the Viewer. (The DIBBs team will not have access to your internal cloud environment).

Thank you!

Reach out to
dibbs@cdc.gov
if you'd like to learn
more!



Disclaimer

For more information, contact CDC
1-800-CDC-INFO (232-4636)
TTY: 1-888-232-6348 [cdc.gov](https://www.cdc.gov)
Follow us on X (Twitter) @CDCgov & @CDCEnvironment

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the U. S. Centers for Disease Control and Prevention.